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AI-Driven Detection of Distracted Drivers to Prevent Road Accidents

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Index

- Abstract
- Problem Statement
- Objectives
- Scope
- Literature Review
- Research Gap
- System Architecture
- Methodology
- Modules
- Algorithms
- Implementation Steps
- Results
- Outputs
- Conclusion
- References

Abstract

Driver distraction is one of the major causes of road accidents worldwide. Traditional monitoring systems and single deep learning models often fail to capture both local visual cues and global contextual dependencies effectively. This project proposes a Hybrid CNN–Transformer architecture for driver distraction classification using a benchmark image dataset. The CNN component extracts local spatial features such as edges and object-level patterns, while the Transformer encoder captures global contextual relationships using self-attention mechanisms. The model is trained and evaluated using standard performance metrics including accuracy, precision, recall, and F1-score. Experimental results demonstrate improved classification performance and better generalization compared to conventional CNN-based approaches.

Problem Statement

Driver distraction causes major accidents, and existing systems lack real-time accuracy, creating the need for a lightweight solution that instantly detects distractions and alerts drivers to prevent crashes.

Objectives

- To develop a deep learning–based driver distraction classification system using a benchmark image dataset.
- To design and implement a Hybrid CNN-Transformer architecture for improved feature representation.
- To preprocess the dataset through resizing, normalization, augmentation, and train-test splitting.
- To train and evaluate Hybrid CNN-Transformer model.
- To compare model performance using accuracy, precision, recall, and F1-score.
- To analyze the effectiveness of integrating CNN-based spatial feature extraction with Transformer-based attention mechanisms for improved distraction classification.

Scope

- The proposed Hybrid CNN–Transformer model can be extended for real-time in-vehicle driver monitoring systems.
- The architecture can be adapted to handle larger and more diverse driver behavior datasets for improved generalization.
- The system can be optimized for lightweight deployment on embedded platforms such as NVIDIA Jetson Nano or edge-based automotive devices.
- The model can be expanded to multi-modal driver monitoring by integrating facial expressions, head pose, eye tracking, and vehicle sensor data.

Literature Review

Paper Title	Authors	Journal	Year	Methods/ Approaches	Key Findings	Limitations
Naturalistic Study of Driver Distraction	Dingus et al.	PNAS	2016	3,500 drivers, naturalistic study	Distractions double crash risk	Not AI-based, observational only
Hybrid CNN-LSTM for Driver Distraction	Jiao et al.	IEEE Transactions	2021	CNN + LSTM	Captures temporal dynamics, better detection of continuous distractions	Heavy architecture, high latency.
DR-MB-SGD Optimized CNN	Qin et al.	Sensors Journal	2022	Synthetic dataset generation	EEfficient training, reduced overfitting, good accuracy	Limited robustness under environmental variations
SynDD1: Synthetic Distracted Driving Dataset	Rahman et al	Computer Vision Conference	2022	CNN multitask (face/hand actions)	Improves model generalization and dataset diversity	May not fully replicate real-world complexities, some distraction types missing

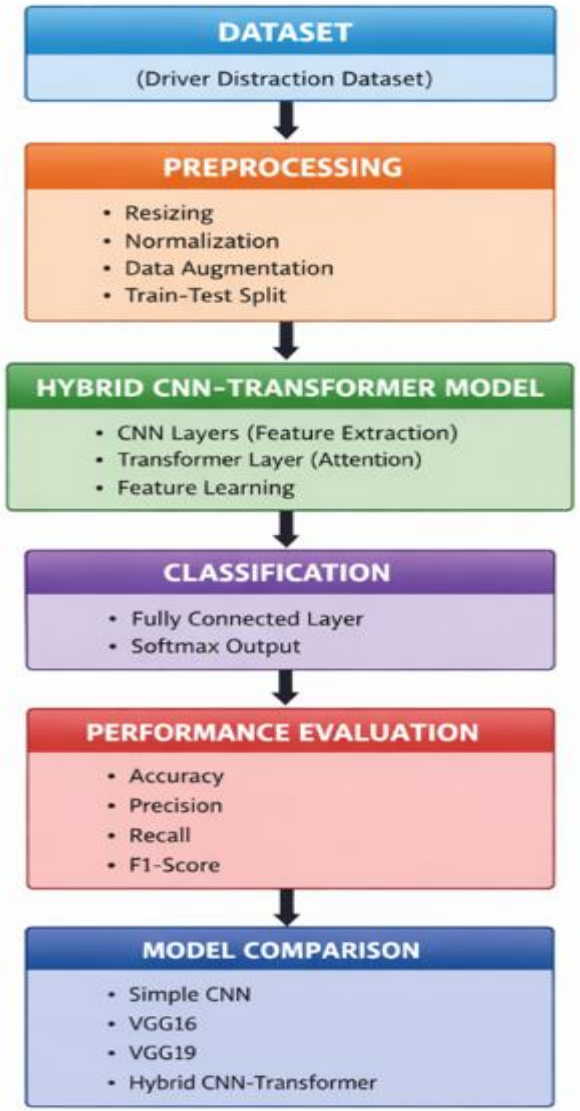
Literature Review

Paper Title	Authors	Journal	Year	Methods/ Approaches	Key Findings	Limitations
Vision Transformer for Driver Distraction	J. Li et al.	Sensors (MDPI)	2022	Vision Transformer (ViT)	Better feature extraction vs CNNs	High compute cost; not edge-suitable
Lightweight CNNs on Edge Devices	Y. Abouelnaga et al.	MDPI / PMC	2022	MobileNet, pruning, quantization	Real-time feasible on Jetson/RPi	Slight degradation under compression
Skeleton-based Distraction Detection	Gong et al.	Computer Vision Conference	2024	OpenPose + KNN	Lightweight, privacy-preserving, real-time on low-power devices	Accuracy depends on pose estimation quality, misses subtle distractions
Integrated Driver & Road Context Detection	Salakapuri et al.	IEEE Transactions	2025	Multi-stream CNN (driver + road)	Holistic monitoring improves robustness	High latency, complex hardware requirement, not practical for embedded deployment

Research Gaps

- **Lack of Lightweight, Deployable Models** : Existing models are too heavy for real-time embedded deployment.
- **Limited Distraction Category Coverage** : Focus mainly on eyes or posture, missing other distraction types.
- **Poor Environmental Robustness**: Fail under low light, glare, or occluded conditions.
- **Insufficient Real-Time Performance**: High latency models hinder instant alerting.
- **Low Interpretability and Driver Trust** : Models give results without explaining reasons to drivers.
- **No Long-Term Validation** : Tested only on short clips, not continuous driving sessions..

System Architecture



Methodology

- Collects benchmark driver distraction dataset containing multiple distraction categories.
- Standardizes image size and normalizes pixel values to ensure uniform model input.
- Applies data augmentation techniques to improve model robustness and reduce overfitting.
- Extracts local spatial features using Convolutional Neural Network (CNN) layers.
- Applies Transformer encoder with multi-head self-attention to model global contextual dependencies.
- Performs feature fusion and classification using fully connected layers.
- Uses Softmax activation for multi-class distraction prediction.
- Evaluates model performance using accuracy, precision, recall, and F1-score.

Modules

- **Dataset Module**
  Handles loading and organizing the driver distraction dataset for training and testing.
- **Preprocessing Module**
  Performs image resizing, normalization, data augmentation, and train-test splitting to prepare data for model training.
- **Feature Extraction Module**
  Uses CNN layers to extract local spatial features from driver images.
- **Attention Learning Module**
  Applies a Transformer encoder to capture global contextual relationships using attention mechanisms.
- **Classification Module**
  Uses a fully connected layer with Softmax activation to classify driver distraction categories.
- **Evaluation and Comparison Module**
  Computes accuracy, precision, recall, and F1-score, and compares the Hybrid CNN-Transformer with Simple CNN, VGG16, and VGG19.

Algorithms

- **Convolutional Neural Network (CNN):**
  Extracts local spatial features such as edges, textures, and object-level details from driver images.
- **Transformer Encoder (Multi-Head Self-Attention):**
  Captures global contextual relationships across feature maps by modeling long-range dependencies.
- **Feature Fusion Strategy:**
  Integrates spatial features from CNN with contextual representations from the Transformer.
- **Fully Connected Layer:**
  Learns complex feature interactions for final classification.
- **Softmax Activation Function:**

  Learns complex feature interactions for final classification.
- **Performance Evaluation Metrics:**
  Accuracy, precision, recall, and F1-score are used to measure and compare model performance.

Implementation

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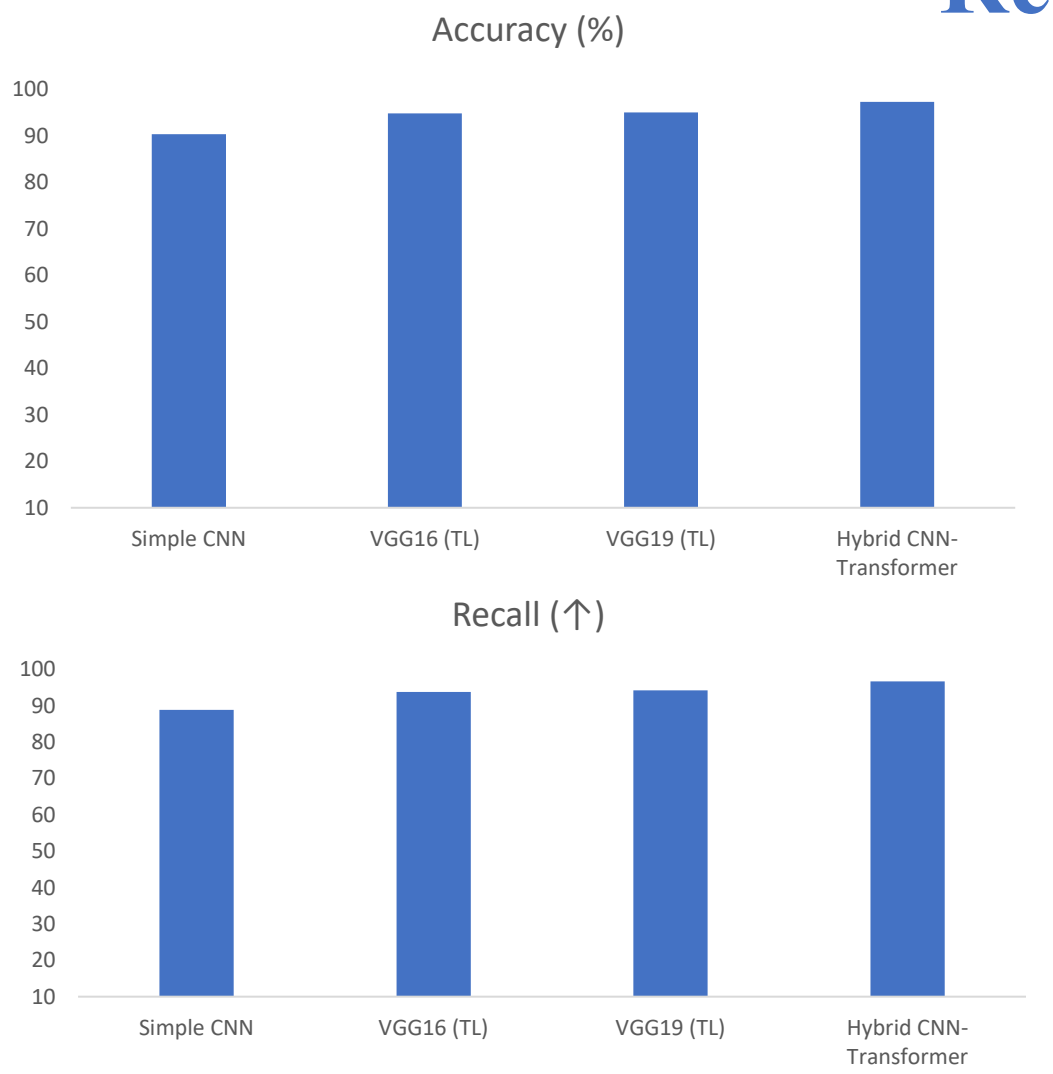
- Uses a benchmark driver distraction dataset organized into multiple distraction categories.
- Images are resized, normalized, and augmented for consistent and robust training.
- CNN layers extract spatial features representing driver posture and object interaction.
- Transformer encoder applies attention mechanisms to capture contextual relationships.
- Extracted features are fused and passed through Dense layers for classification.
- Softmax layer generates probability scores for each distraction class.
- Model performance is evaluated using classification metrics on test data.

Results



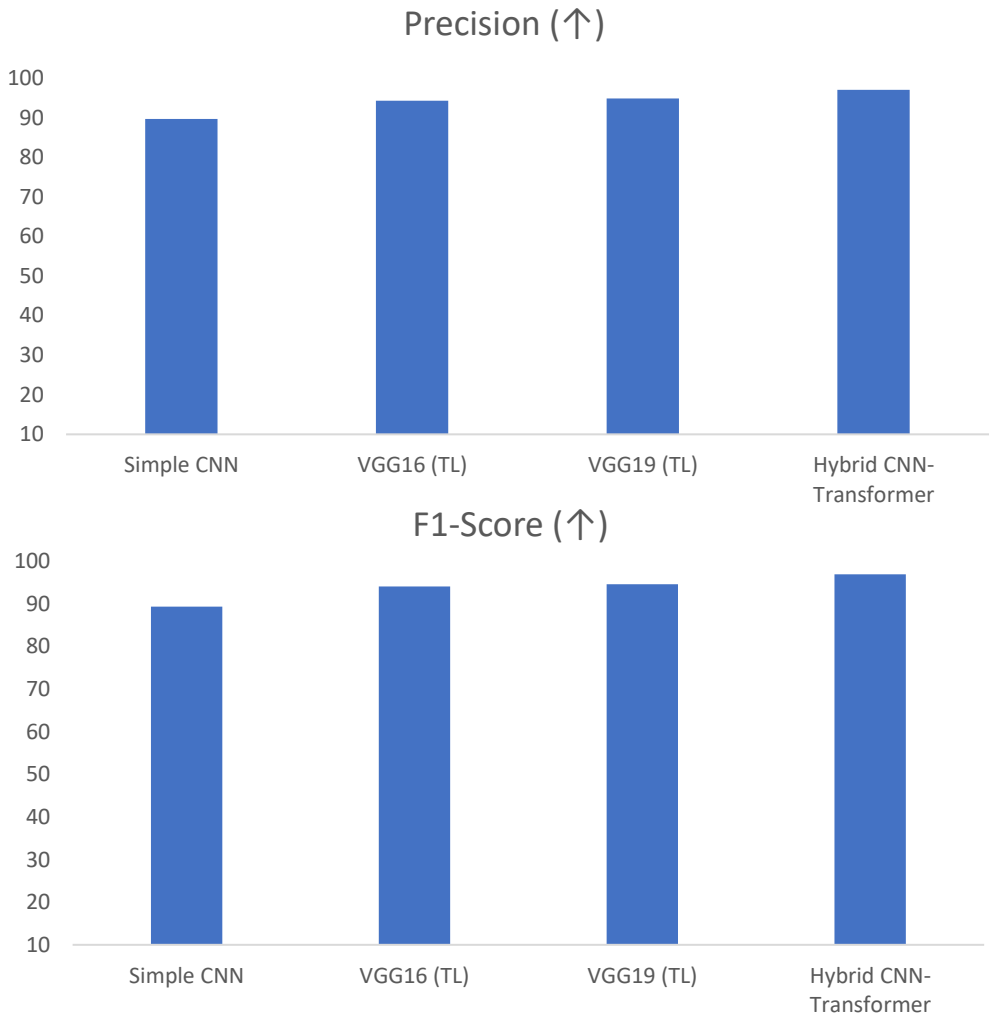


Fig :Performance Comparison of Models

Output

📊 SimpleCNN – Test Performance

	precision	recall	f1-score	support
c0	0.000	0.000	0.000	200
c1	0.000	0.000	0.000	200
c2	0.000	0.000	0.000	200
c3	0.000	0.000	0.000	200
c4	0.000	0.000	0.000	200
c5	0.000	0.000	0.000	200
c6	0.000	0.000	0.000	200
c7	0.100	1.000	0.182	200
c8	0.000	0.000	0.000	200
c9	0.000	0.000	0.000	200
accuracy			0.100	2000
macro avg	0.010	0.100	0.018	2000
weighted avg	0.010	0.100	0.018	2000

```
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: U
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: U
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: U
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

📊 VGG16 – Test Performance

	precision	recall	f1-score	support
c0	0.915	0.915	0.915	200
c1	0.970	0.980	0.975	200
c2	0.950	0.945	0.947	200
c3	0.819	0.885	0.851	200
c4	0.832	0.770	0.800	200
c5	0.995	0.975	0.985	200
c6	0.918	0.955	0.936	200
c7	0.990	0.995	0.993	200
c8	0.885	0.850	0.867	200
c9	0.925	0.930	0.928	200
accuracy			0.920	2000
macro avg	0.920	0.920	0.920	2000
weighted avg	0.920	0.920	0.920	2000

Output

📊 VGG19 – Test Performance

	precision	recall	f1-score	support
c0	0.000	0.000	0.000	200
c1	0.000	0.000	0.000	200
c2	0.000	0.000	0.000	200
c3	0.000	0.000	0.000	200
c4	0.000	0.000	0.000	200
c5	0.105	0.925	0.189	200
c6	0.109	0.130	0.119	200
c7	0.000	0.000	0.000	200
c8	0.000	0.000	0.000	200
c9	0.000	0.000	0.000	200
accuracy			0.105	2000
macro avg	0.021	0.106	0.031	2000
weighted avg	0.021	0.105	0.031	2000

```
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

📊 Hybrid – Test Performance

	precision	recall	f1-score	support
c0	0.985	0.980	0.982	200
c1	0.995	0.995	0.995	200
c2	1.000	0.990	0.995	200
c3	0.990	0.985	0.987	200
c4	0.970	0.980	0.975	200
c5	0.995	0.990	0.992	200
c6	0.966	1.000	0.983	200
c7	0.990	1.000	0.995	200
c8	0.990	0.965	0.977	200
c9	0.985	0.980	0.982	200
accuracy			0.987	2000
macro avg	0.987	0.987	0.987	2000
weighted avg	0.987	0.987	0.987	2000

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Conclusion

The proposed Hybrid CNN–Transformer architecture enhances driver distraction detection by integrating spatial feature extraction with attention-based contextual modeling. The combination of convolutional layers and Transformer mechanisms improves feature representation and classification accuracy. Experimental evaluation demonstrates strong performance and better generalization across distraction categories. Although challenges remain in real-time deployment and environmental robustness, the hybrid approach shows significant potential for intelligent driver monitoring systems and advanced driver assistance technologies.

Thank you!

Paper Details

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